so (E_K and) v increases A1 **[2]**

2 **(a)** e.g. fixed mass/ amount of gas
ideal gas
(*any two, 1 each)* B2 **[2]**

(b) **(i)** $n = pV / RT$ C1
$= (2.5 \times 10^7 \times 4.00 \times 10^4 \text{ x } 10^{-6}) / (8.31 \times 290)$ C1
$= 415$ mol A1 **[3]**

(ii) volume of gas at 1.85×10^5 Pa $= (2.5 \times 10^7 \times 4.00 \times 10^4) / (1.85 \times 10^5)$
$= 5.41 \times 10^6$ cm^3 C1
so, $5.41 \times 10^6 = 4.00 \times 10^4 + 7.24 \times 10^3 N$ C1
$N = 741$ A1 **[3]**
(answer 740 or fails to allow for gas in cylinder, max 2/3)

3 (a) gradient of graph is (a measure of) the sensitivity M1